Cardinality of a set X, denoted by |X| *or* #X, is the number of distinct elements in the set **X**. For example,

$|\{1, 3, 7\}| = 3$, $|\varnothing| = 0$; $|N| = \infty = |Z| = |Q|$, $|\{0\}| = 1$

Two unequal/ different sets may have same/ equal cardinality. For example,

$\{1, 3, 5\} \neq \{1, 3, 7\}$, but their cardinalities are equal, as $|\{1, 3, 5\}| = |\{1, 3, 7\}| = 3$.

Check Your Progress 1

1) Which of the following collections are sets? Justify your answer.

   i) The collection of all the months of a year beginning with the letter M.

   ii) The collection of all the months of a year beginning with the letter Z.

   iii) The collection of ten most talented living writers of the world.

   iv) A team of eleven most renowned football players of the world.

   v) $\{1, 2, 2, 3\}$

   vi) $\{1, 2, \{2, 3\}, 3\}$

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2) For each of the sets consisting of following members, write it in roster/ tabular form and its cardinality:

   a) positive prime integer factors of 180

   b) solutions of the quadratic equation $x^2 - 3x - 10 = 0$

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3) Tell which of the following is a set, and if it is a set, then (i) write it in set-builder form (ii) tell its cardinality.

   a) $\{6, 9, 12, 15, 18, 21, 24, 27\}$

   b) $\{2, 4, 8, 16, \ldots\}$

   c) $\{\{1\}, \{1, 2\}, \{1, 2, 3\}, \ldots, \{1, 2, 3, 4 \ldots 10\}\}$

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1.2.4 Relationships between Sets

For any two sets X and Y, the set relationships that we discuss are (i) *equality (=) of X and Y* (ii) X is a *subset/ superset(⊆)* of Y (iii) X is a proper *subset/superset(⊂)*of Y, and (iv) is *not a subset* of (**'⊄').**

1.2.4.1 Notations, Definitions and Examples

i) Two sets X and Y are said/defined as ***equal*** if every element of X is an element of Y, *and* every element of Y is an element of X.

   The n*otation X= Y* is used for 'equality of sets X and Y'

   Example: Let X = {1, 2, 3, 4, 5}, and Y= {x: x is a natural number such that 1≤ x≤5}, and W= {3, 5, 2, 4, 1}. Then X = Y = W.

ii) X is said to be a ***subset*** of Y if every element of X is an element of Y. Instead of saying '***X is a subset of Y***' we may equivalently say ***'Y is a superset of X', or 'X is contained in Y'*** or '***Y contains X***'.

   The notation X⊆ Y is used for 'X is a subset of Y'.

   Examples: ***(i)*** Let X = {a, e, i}, and Y = set of all vowels in English language. Then X⊆ Y. (ii) Every set is its own subset, i. e.X ⊆ X (can be easily verified by applying the definition).

iii) X is said to be a ***proper subset*** of Y if (i) every element of X is an element of Y, *and also*(ii) there is at least one element of Y which does not belong to X. Then, it can also be said that Y is a proper superset of X.

   The notation X ⊂ Y is used for 'X is a proper subset of Y' and also for 'Y is a proper superset of X'.

Example: Let X = set of even decimal digits = {0, 2, 4, 6, 8} and let Y= set of all decimal digits = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}. Then **X⊂ Y, as every even decimal digit is also a digit. But 1**∈ Y, and 1**∉ X.**

The notation **X ⊄ Y** denotes the fact that X is ***not a subset*** of Y. For showing X ⊄ Y, we need to show *only* that there is some *one* element of X which does not belong to Y. For example,Let X = {1, 5, 10, 15}, and Y = {1, 12, 14, 15}. To show **X ⊄ Y**, we need *only* to say/show that 5 ∈ X, and 5 ∉ Y. We *need* not go further to say that 10 ∈ X, and 10 ∉ Y. *Also, X ⊄ X, as condition (ii) for being proper subset is not satisfied.*

Remarks: There is **no complete standardization of the notation of subset**. Some people use ⊂ Y, as every even decimal digit is also a digit. But 1 '⊂' to denote a subset, which may or may not be a proper subset. Then, they use the symbol ⊊ for proper subset. We could have used these symbols. But we would continue with symbols '⊆' and '⊂' denoting respectively 'subset' and 'proper subset'.

1.2.4.2 Establishing Set Relationships

Here, through a simple example, we demonstrate how to establish each of these set relationships. Later, we will consider some more complex examples of establishing these relationships, particularly regarding the equality (=).

Let us consider sets

$A = \{x: x^2+3x + 2 = 0\}$, the set of solutions of the equation $x^2 + 3x + 2 = 0$.

$B = \{ -1, -2\}$, $C= \{ -2\}$, $E = \{0, -1, -2\}$

i) Easiest case is that of ***'is not a subset of':*** In our case**,**

$\boldsymbol{E \not\subset C}$**,** as $0 \in E$, *but* $0 \notin C$**.** Similarly, $\boldsymbol{E \not\subset A}$***, as*** $0 \in E$, *but 0 is not a root of* $x^2+3x + 2 = 0$.

ii) Next, in order to show $B \subseteq E$, we need to consider, one by one, each element of B, and show that the considered element belongs to E also. In complex cases, we do not consider each element one by one, but take an arbitrary element. The methodology will be explained later.

iii) Further, in order to show that B is a ***proper subset***of E (i. e.$B \subset E$), in addition to what we have done in (ii) above, we need to show the existence of an element of E, which does not belong to B. In our case $0 \in E$, but $0 \notin B$.

iv) To show equality (=) of two sets, we need to repeat case (ii) two times, reversing the roles of the sets under consideration.

In our case, A= B, because -1, which belongs to B, is a root of $x^2+3x + 2 = 0$, hence, belongs to A. Similarly, -2, which belongs to B, is a root of $x^2+3x + 2 = 0$, hence, belongs to A. Conversely, as $x^2+3x + 2 = 0$ is a quadratic equation, it has exactly two roots, viz. $-1, -2$.

Check Your Progress 2

1) For the pair of sets X and Y, state whether X = Y or not, along with explanation for your response.

   i) $X = \{2, 4, 6, 8, 10\}$, $Y = \{x: x$ is positive even integer and $x \leq 10\}$

   ii) $X = \{x: x$ is a multiple of $10\}$, $Y = \{10, 15, 20, 25, 30 \ldots\}$

   iii) $X = \{2, 3\}$, $Y = \{x: x$ is solution of $x^2 + 5x + 6 = 0\}$

   iv) X= {x: x is a letter in the word FOLLOW}, Y = {y: y is a letter in the word WOLF}

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2) Find out the correct relation of $X \subseteq Y$, $Y \subseteq X$, $X \subset Y$, $Y \subset X$ or $X \not\subset Y$ etc for the following pairs of sets X and Y:

   i) X= {x: x is a student enrolled in IGNOU and having Economics as one of the courses, Y= {x: x is a student enrolled in IGNOU}
   ii) X= {x: x is a triangle in a plane}, Y= {x: x is a rectangle in the plane}
   iii) X= {x: x is an even natural number}, Y= {x: x is an integer}
   iv) X = {1,2,3} and Y = {1, {2,3}, 4, 5}

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3) For X = {1, {2, 3}, 3, 4}, which ones of the following is/are true statement(s)?

   (i) $2 \in X$ (ii) $\emptyset \in X$ (iii) $\emptyset \not\subset X$ (iv)$\emptyset \subset X$(v) $\{2,3\} \in X$ (vi) $\{2,3\} \subset X$, (vii) $\{\{2,3\}\} \subset X$

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1.2.5 Some Special Sets: Universal Set, Null Set, Power Set and Convex Set

Next, we briefly discuss some of the special type of sets, which we encounter frequently in Mathematics, especially relating to economic problem solving.

a) **Universal Set**: For discussing problems relating to educating people in India, we need not take into consideration all human beings, we need to consider *set of only Indians*. While using mathematics for solving problems (like educating Indians) from some domain of human experience, some sets of entities from the domain are considered as and when required. For facilitating arguments, a *sufficiently large set of entities* from the problem domain— *which includes all possible entities* that may be required at any time during the discussion or solution-process of solving the problems of the type under consideration—is taken. For example, *in our present case, the set of all Indians.* ***Such a set is called the Universal set and is generally denoted by U.*** *Thus, by definition, every set X under consideration will be a subset of the Universal set U. i. e.* ***$X \subseteq U$ for all sets X under consideration during solving a particular (type of) problem.***

Remarks:

- It may be noted that for different problem-types or different contexts, there will be different universal sets.
- By definition of universal set U, every set X is a subset of U, i. e. $X \subseteq U$. Also, from the fact that $X \subseteq X$ for every X, we get $U \subseteq U$.

- The concept of '*Universal set*', since its inception, has been both controversial and problematic. Specially, it has led to a number of well-known paradoxes including the Russell's paradox, which implies that X, the collection of all sets, is not a well-defined collection, and hence, X is not a set.

However, the concept of *Universal Set* has facilitated representation and solution of problems from most of the domains of our concern including from mathematics itself.

b) **The empty or null set:** It may appear strange, at least when heard for the first time, that a set may have no member at all, or the number of elements in a set may be zero. According to our definition of a set, the following is a set

$$S = \{x: x \text{ is an even integer, } \textit{and } x^2 = 9\}.$$

The collection S is well-defined, and hence, is a set. But there is no member which satisfies the conditions of its membership. Therefore, the number of members of S is zero. There are many combinations of conditions, which may make a set to have no member**.** Each of such sets is called a *Null set, a Void set or an Empty set****.*** ***A null set is denoted by the standard notation*** Ø or by {}.

Remarks: Ø is a subset of every set X, i. e. Ø⊆ X, for every set X.

The fact is easily established through contradiction, a common proof technique in Mathematics: Assume what is to be proved as wrong, and then reach a fact which conflicts with some earlier established fact. So, let Ø⊆ X be false. Then there must an $x \in Ø$, *and* $x \notin X$. But $x \in Ø$ is not possible by definition of Ø. Hence, the assumption that 'Ø⊆X be false' is wrong. Hence the proof.

Proper subsets of a set X: As Ø and a set X itself are subsets of the set X. These two subsets are not considered as proper subsets. All other subsets are considered as proper subsets.

c) **Power Set of a Set:** The set of all subsets of a given set A is called Power Set of A, and is denoted as ***P(A)***. For example, if A= {1, 2, 3}, then P(A) = { ∅, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3}}. Sometimes power set of a given set A is denoted by 2^A, because that is the cardinality of the power set A.

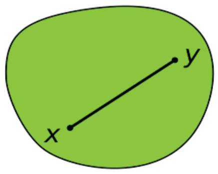
Next, we state, without proofs, some important results about the relation of *subset*

- $A \subseteq A$ for every set A. Every set is a subset of itself.
- If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$. We say that ⊆ is a *transitive relation* between sets.
- If $A \subseteq B$ and $B \subseteq A$ then $A = B$.
- If $|A|$= n, then P(A) = 2^n. For example, the set $\{1, 2, 3\}$ has 3 elements,

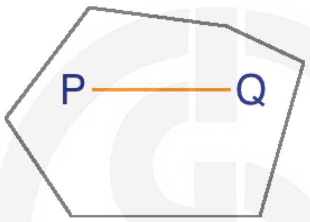
and, the power set of A, viz. P(A) = {∅, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3}} has $2^3 = 8$ elements.

Next, we discuss a type of set which is frequently encountered while attempting solutions of economy related problems.

d) **Convex Set:** A convex set in a space (generally 2-dimensional real-space) is a **set** of points such that, given any two points A, B in the **set**, the line AB joining them lies entirely within that **set.**



Thus, a subset S of a space is called *convex* if and only if for any pair of points p and q in the set S, the line section from p to q is enclosed completely in the set of S, as done in the figure on L.H.S. below.



Convex

Non-Convex

Algebraically, if x_1 and x_2 are two numbers in the set S, then all the points on the straight line $x_1 x_2$—which can be represented as $\lambda x_1 + (1 - \lambda)x_2$, for any λ such that $0 \leq \lambda \leq 1$—also belong to S.

Check Your Progress 3

1) Tell which one of the following is a convex set, within set of real numbers.

   (i) [4, 5[= {x: 4 ≤ x < 5}, (ii) [1, 4] ∪ [7, 11[= {x: 1 ≤ x ≤ 4} ∪ {x: 7≤ x < 11}

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2) What is the Cardinality of the Power set of the set {0, 1, 2}?

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3) Let $U = \{u, v, w, x, y, z\}$.

   i) Find the number of subsets of U.

   ii) Find the number of proper non-empty subsets of U.

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4) List the 8 subsets of $\{a, b, c, d\}$ containing $\{d\}$.

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1.2.6 Quantifiers and Logical Symbols/ Operators

Quantifiers and Logical operators appear frequently throughout mathematics.

a) **Quantifiers**

$(\forall x)\ x \in X$ denotes *'for all $x \in X$';*

$(\exists x)\ x \in X$ denotes *'for some $x \in X$';*

$(\nexists x \in X)$ *'there does **not** exist an x in X'* OR 'there is ***no*** x in X, which satisfies'.

Brief explanation of Quantifiers: The phrases like *'for all'* and *'for at least one'* occur frequently in the discussion of properties of elements of a given set. These phrases are called *quantifier phrases*. The following special symbols have been introduced to economize on number of letters used to represent these phrases. Let X be a given set, and x be an arbitrary element of X, then

$(\forall x)$ denotes the phrase *'for all $x \in X$'.*

It may be noted that the set X is understood to be given or known already for the notation. However, if the set under consideration is not clear, then the name of the set also may be included in the notation as:

$(\forall x \in X)$ or $(\forall x)\ x \in X$

For example, for the set *Sq_int = {0, 1, 4, 9,…},* it is true that every one of its members is square of some integer. This fact may be denoted as

$(\forall x \in Sq\_int)\ (x = y^2)$, *for some* $y \in Z$

Another frequently used quantifier phrase is *'for some'*. In mathematics and logic, *'for some'* is taken as equivalent of *'for at least one'*.

For explanation again, let X be a given set, and x be an arbitrary element of X, then

$(\exists x)$, or $(\exists x \in X)$ denotes the phrase *'for some $x \in X$'*.

For example, for the set *Sq_int = {0, 1, 4, 9 …},* it is true that some element of *Sq_int* is greater than 100 (because, 144 and many other members of *Sq_int* are greater than 100). The fact may be denoted as

$(\exists x \in Sq\_int)\ (x > 100)$.

The notation for the earlier statement

$(\forall x \in Sq\_int)\ (x = y^2)$, *for some* $y \in Z$

may be further refined as

$(\forall x \in Sq\_int)\ (\exists y \in Z)\ (x = y^2)$, Z is the set of all integers.

Sometimes, we need to use the phrase *'there is no x in X'* or equivalently, *'there does not exist an x in X'* etc. Such a phrase is denoted as $(\nexists x \in X)$.

For example, for the set $Sq\_int = \{0, 1, 4, 9\ldots\}$, it is known that *no* member of *Sq_int* is less than zero. The fact may be denoted as

$(\nexists x \in Sq\_int)\ (x< 0)$.

b) **Logical operators**

$P \rightarrow Q$ denotes '*If P then Q',*

'$P \leftrightarrow Q$' denotes 'P *if and only if Q',*

'$P \leftrightarrow Q$' is equivalent to both $P \rightarrow Q$ *and* $Q \rightarrow P$ combined.

Brief explanation of Logical operators:

The *logical* expressions of the form '*If P then Q'* occur frequently, where, P and Q are statements (i. e. each of P and Q is either 'true' or 'false', and not both 'true' and 'false' simultaneously). The logical expression '*If P then Q'* is denoted by $P \rightarrow Q$.

For example, the statement *'If x is an integer, then its square is a non-negative integer'* may be denoted as $(\forall x \in Z) \rightarrow (x^2 \geq 0)$

Also, the logical expression 'P *if and only if Q'* occurs frequently in mathematical arguments. The expression 'P *if and only if Q' is sometime shortened as* 'P ***iff*** *Q'*. The expression is denoted by

'$P \leftrightarrow Q$'

The notation $P \leftrightarrow Q$ is equivalent to both '$P \rightarrow Q$ and $Q \rightarrow P$' taken together.

For example, we may say that '$x^2 = 100$ *if and only if* $x =10$ *or* $x = -10$'.

The notation for this true statement is: $(x^2 = 100 \leftrightarrow (x =10 \text{ or } x = -10))$.

Further, it may be noted that though $((x =10) \rightarrow (x^2 = 100))$ is true, *yet* $(x^2 = 100 \rightarrow x =10)$ is ***not*** (essentially) true. Because, $x^2 = 100$ *then x may be* (-10) *also*. Hence, to say $(x^2 = 100 \rightarrow x =10)$ is INCORRECT. But to say $(x = 10 \rightarrow x^2 =100)$ is CORRECT.

1.2.7 Some Conceptual Clarifications and Explanations

In this subsection, we discuss the difference between the concepts of 'Relation' & 'Operation', Sets operations: union, intersection, difference, complementation, more clarifications: operands/ argument/ inputs; value/ result/ output; operation vs. operand.

1.2.7.1 Difference between the Concepts of 'Relation' and 'Operation'

Let us now understand the difference between the two concepts '*relation*' and '*operation'*, which is illustrated through the following examples for numbers: for numbers 8 and 15, the notation 8 < 15 denotes a *relation* 'is less than'; similarly, the notation 8 ≠ 15 denotes the relation '*is not equal to*', and 3+5 = 8 denotes the relation *'is equal to'*. Each of the relations does NOT give a new number, but, tells some already existing fact, e. g. the notation 8 ≠ 15 denotes the already existing fact of 8 'not being equal to' 15.

A relation returns a value TRUE or FALSE. For example:

3< 5 returns TRUE; 7< 5 returns FALSE.

8 ≠ 15 returns TRUE; 8 ≠ 8 returns FALSE.

On the other hand, *an **operation** gives something new;* in the following cases new numbers: 8+15 gives new number 23; 8 – 15 gives new number – 7; and (+√**16)** gives a new number 4.

To apply the operation of '+', we must have at least two numbers, therefore, '+' is called a **binary operation.** On the other hand, to take a square-root (√**),** we need only one number, therefore, operations like square-root (√**)** is called **unary operation.**

For discussion in general, we need the concept of cross-product of sets.

Cross-Product of sets: for two non-empty sets A_1 and A_2, their cross-product (Cartesian product), denoted as $A_1 \times A_2$, is the set of all ordered pairs (a_1, a_2), where $a_1 \in A1$and $a_2 \in A_2$. The concept may be generalized to any finite number of sets say $A_1, A_2, \ldots A_n$., denoted as $A_1 \times A_2 \times \ldots \times A_n$. ***The cross-product of n sets*** is the set of all n-tuples of the form (a_1, a_2. …, a_n), where $a_i \in$ Ai for I =1, 2, ... n.

Relation on a set X is a subset of cross-product of the X with itself n times, where *n* is the arity of the Relation. For example, *is-male* involves a single person, hence, it is a relation of arity one; *is-mother-of* involves two persons, hence its arity is two. A relation *returns as value* exactly one of 'true' or 'false'. For example, if Sheela is not mother of Mohan, then the value of the relational expression *is-mother-of (Sheela, Mohan)* returns false.

Operation on a set X is quite similar to the concept of *relation*. An operation on X maps the cross-product of X with itself n times, where *n* is the arity of the operation. But it *returns, as value, an element of X itself*. For example, operation of *square on N,* the set of natural numbers, involves a single number, hence, it is an operation of arity one; the operation of *sum* involves two numbers, hence its arity is two. But in both the cases of *square* and of *sum*, the value returned is a single number/ member of X.

Connection between relation on a set and operation on a set:

An *(n+!)-ary relation* on a set can be converted to an *n-ary operation* on the set. For example, *is-mother-of* is a binary relation on the set X of all human beings. This binary relation can be converted to unary operation *name-mother-of* on X. which returns the name of the mother of x ∈X. For example, if Sita is mother of Mohan, then the binary relation *is-mother-of (Mohan, Sita)* is true. On the other hand, the unary operation *name-mother-of (Mohan)* returns Sita.

1.2.7.2Operations *on* Sets: Union, Intersection, Difference, Complementation

We have already discussed Relations between sets. Now we discuss Operations *on* Sets. Similar to the case of numbers, for the sets also, we can get new sets from given sets by employing operations. Here we discuss only three *binary set operations*, viz. *union, intersection and difference of sets*. By *binary,* we mean, each of these operations requires TWO sets for its application. Further, we discuss one ***unary*** set operation viz. *complementation*, i. e. complementation requires ***only one*** set for its application.

a) ***Union X∪Y of two sets X and Y is defined as***

$X \cup Y = \{x: x \in X \textbf{ or } x \in Y\}$

Here, x ∈ X **or** x ∈ Y *also includes the case that x may belong to both of X and Y.*

For example, if X= {1, 2, 3, 4} and Y = {2, 3, 5, 8, 9}, then

$X \cup Y = \{1, 2, 3, 4, 5, 8, 9\}$

Elements, like 2 and 3, which are occurring in both X and Y appear only once in $X \cup Y$, as elements are not repeated in representation of a set.

b) ***The Intersection*** $X \cap Y$ ***of two sets X and Y is defined as***

$X \cap Y = \{x: x \in X \textbf{ and } x \in Y\}$

For example, if X= {1, 2, 3, 4} and Y = {2, 5, 8, 3, 9}, then $X \cap Y = \{2, 3\}$,

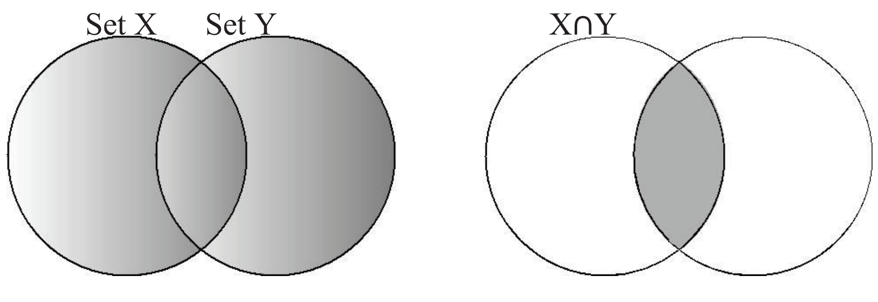

Fig. 1.1: On L.H.S., shaded area is the union $X \cup Y$; and on R.H.S., the shaded area is the intersection $X \cap Y$ of the sets X and Y.

c) ***The Difference X – Y of two sets X and Y is defined as***

$X - Y = \{x: x \in X \text{ and } x \notin Y\}$